Diskretna vremenski oniska
FOURIEROVA TRANSF.

DTFT $\quad F(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} f[n] \cdot e^{-j\Omega n}$

IDTFT $\quad f[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} F(e^{j\Omega}) e^{j\Omega n} \, d\Omega$

Kod Fourierovih redova smo računali snagu, ovdje radimo s energijom

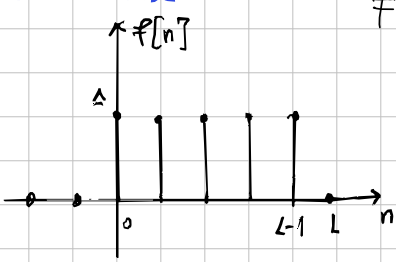
Dokaz periodičnosti:

$$F(e^{j(\Omega+2\pi k)}) = \sum_{n=-\infty}^{\infty} f[n] e^{-j(\Omega+2\pi k)n} = \sum_{n=-\infty}^{\infty} f[n] \cdot e^{-j\Omega n} \cdot \underbrace{e^{-j2\pi k n}}_{1}$$

$$F(e^{j(\Omega+2\pi k)}) = \sum_{n=-\infty}^{\infty} f[n] \cdot e^{-j\Omega n} = F(e^{j\Omega}) \quad \checkmark$$

$$E = \sum_{n=-\infty}^{\infty} f[n] \cdot f^*[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} F(e^{j\omega}) \cdot F^*(e^{j\omega}) \, d\omega = \frac{1}{2\pi} \int_{-\pi}^{\pi} |F(e^{j\Omega})|^2 \, d\Omega$$

Zadatak:



$$F(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} f[n] \cdot e^{-j\Omega n} = \sum_{n=0}^{L-1} A \cdot e^{-j\Omega n} = A \underbrace{\sum_{n=0}^{L-1} (e^{-j\Omega})^n}_{\text{geom.}}$$

$$= A \frac{1-(e^{-j\Omega})^L}{1-e^{-j\Omega}}$$

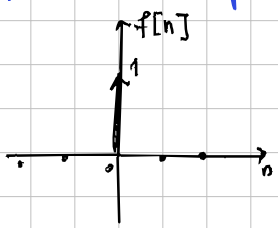
$$= A \frac{e^{-j\Omega \cdot 0} - e^{-j\Omega L}}{e^{-j\Omega \cdot 0} - e^{-j\Omega}} = A \cdot \frac{e^{-j\Omega \frac{L}{2}} \left(e^{j\Omega \frac{L}{2}} - e^{-j\Omega \frac{L}{2}}\right)}{e^{-j\Omega \cdot \frac{1}{2}} \left(e^{j\Omega \frac{1}{2}} - e^{-j\Omega \frac{1}{2}}\right)}$$

$$= A \cdot e^{-j\Omega\left(\frac{1}{2} - \frac{L}{2}\right)} \cdot \frac{\cancel{(-2j)} \sin\left(\Omega \frac{L}{2}\right)}{\cancel{(-2j)} \sin\left(\Omega \frac{1}{2}\right)}$$

$$\boxed{= A \cdot e^{-j\Omega\left(\frac{L}{2} - \frac{1}{2}\right)} \cdot \frac{\sin\left(\Omega \frac{L}{2}\right)}{\sin\left(\Omega \cdot \frac{1}{2}\right)}}$$

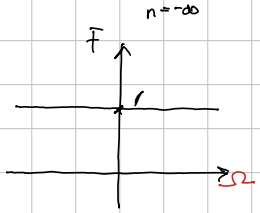
$$F(e^{j \cdot 0}) = A \sum_{n=0}^{L-1} \underbrace{e^{-j0 \cdot n}}_{1} \Rightarrow \boxed{F(e^{j \cdot 0}) = AL}$$

Zadatak: Impuls



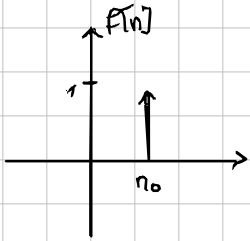
$$F(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} f[n] e^{-j\Omega n} = \sum_{n=-\infty}^{\infty} \delta[n] (e^{-j\Omega})^n = 1$$

$n = 0 \rightarrow$ impuls



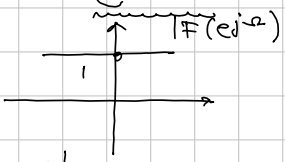
Zadatak:

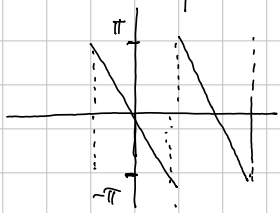
$$F(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} \underbrace{\delta[n-n_0]}_{n=n_0} e^{-j\Omega n} = e^{-j\Omega n_0}$$



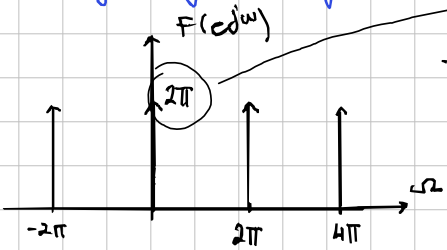
$$|F(e^{j\Omega})| = 1$$

$$\measuredangle F(e^{j\Omega}) = -n_0 \cdot \Omega$$



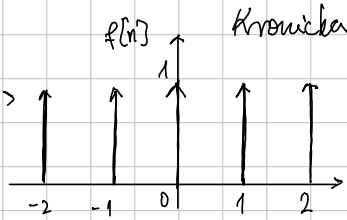


i ovdje vrijedi svojstvo dualnosti



$$\rightarrow f[n] = \frac{1}{\cancel{2\pi}} \int_{-\pi}^{\pi} \cancel{2\pi} \cdot \underbrace{\delta[n]}_{n=0} \overset{=1}{\boxed{e^{j\Omega n}}} d\Omega$$

kada je impuls onda „ne" rješavamo integral

$f[n] = 1 \rightarrow$ to je konstanta, ALI konstanta Kroneckerovih impulsa

$\Longrightarrow$